

Specifications

Specimen Systems

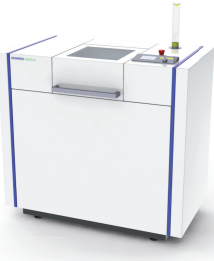
Peak Energy
Variable Spotsize
Filter Changer (3 pos.)
Max. Scan Diameter
Max. Scan Length/Height
Max. Sample Diameter/Length
Automatic Sample Changer Option
Resolution (Nom. /10% MTF)
Image Matrix : from
to

μCT 45 ¹ NEW	μCT 90 ¹ NEW
30-90 kVp 5/7 μm (@4/8W) yes 50 mm 120 mm 90 mm / 120 mm up to 20 holders 1.25-90 μm / <5 μm 512 x 512 8192 x 8192	30-90 kVp 5/7 μm (@4/8W) yes 90 mm 120 mm 90 mm / 120 mm up to 20 holders 1.75-180 μm / <9 μm 512 x 512 8192 x 8192



Peak Energy
Variable Spotsize
Filter Changer (4 pos.)
Max. Scan Diameter
Max. Scan Length/Height
Max. Sample Diameter/Height
Automatic Sample Changer Option
Resolution (Nom. /10% MTF)
Image Matrix : from
to

μCT 50	μCT 100	μCT 100 HE
30-100 kVp 4-30 μm (4-18W) yes 50 mm 120 mm 100/160 mm up to 12 holders 0.5-100 μm / <2 μm 512 x 512 8192 x 8192	30-100 kVp 4-30 μm (4-18W) yes 100 mm 140 mm 100/160 mm up to 12 holders 1.25-200 μm / <4 μm 512 x 512 8192 x 8192	30 - 130 kVp 5 - 40 μm (4-39W) yes 95 mm 140 mm 100 /160mm up to 12 holders 1.25-200 μm / <5 μm 512 x 512 8192 x 8192



in vivo Systems

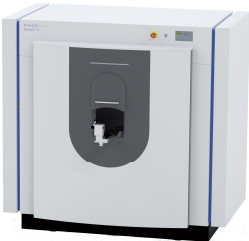
Peak Energy
Max. Scan Diameter
Max. Scan Length
Resolution (Nom./10% MTF)
Minimal Scan Time per Stack
Image Matrix : from
to

vivaCT 80
30-70 kVp 80 mm 145 mm 5 - 160 μm / <14 μm 20 s 512 x 512 8192 x 8192



Peak Energy
Max. Scan Diameter
Max. Scan Length
Resolution (Nom./10% MTF)
Standard Scan Time per Patient
Image Matrix : from
to

XtremeCT II
68 kVp 140 mm 200 mm 17 μm / <55 μm 1.6 min 512 x 512 8192 x 8192



Specialized specimen holders and/or casts, quality control protocols
Density and resolution phantoms available for all scanners (some are optional)

Computing

HP Integrity Server, 64 bit

rx2800 i4/i6	bl860c/bl870c/bl890c i4/6
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RAM	up to 768 GB	up to 1.5 TB
CPUs/cores	1-2/4-16	1-8 /4-64
Disks	4 TB to multiple TB	4 TB to multiple TB
Archival Device	high capacity tape drive (15.0 TB per cartridge)	
Reconstruction Time	2 s for 1k x 1k (1 CPU)	2 s for 1k x 1k (1 CPU)
Analysis Time	5 min typical	



Pictures are not to scale

Clustering of multiple rx2800/bl8x0 possible to enhance throughput
GPU Reconstruction to speed up reconstruction with up to 120 times, runs on NVIDIA GPU Accelerator with Pascal architecture (Px000)
¹ Specifications subject to change ² Contact Scanco Medical for latest update on regulatory approval in your country

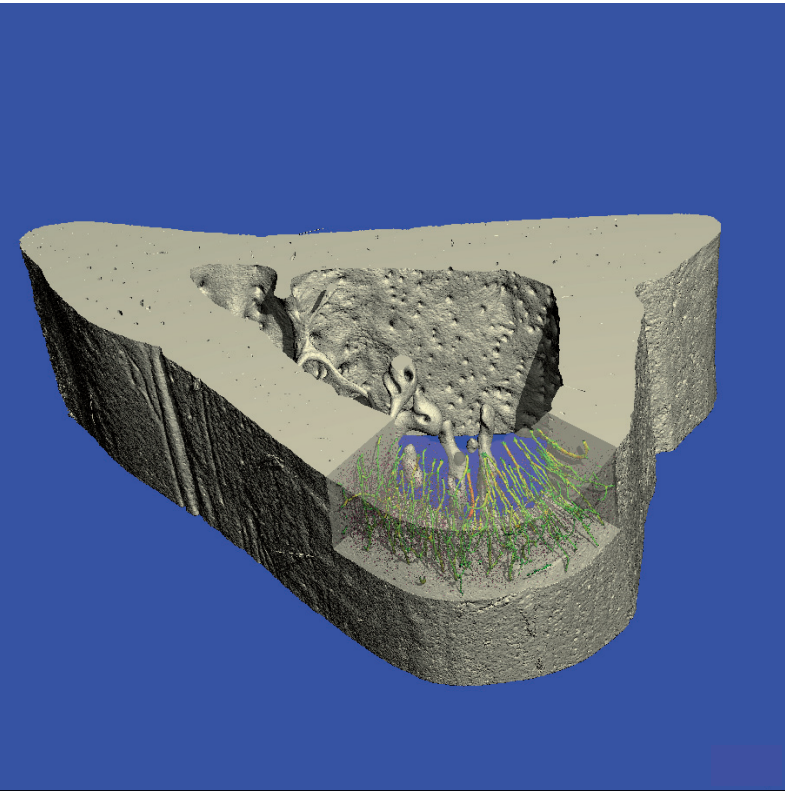
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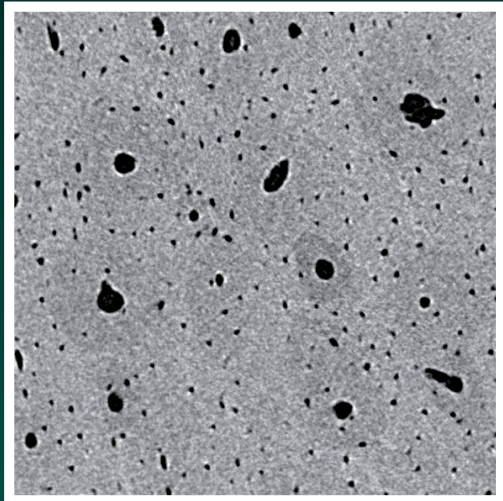
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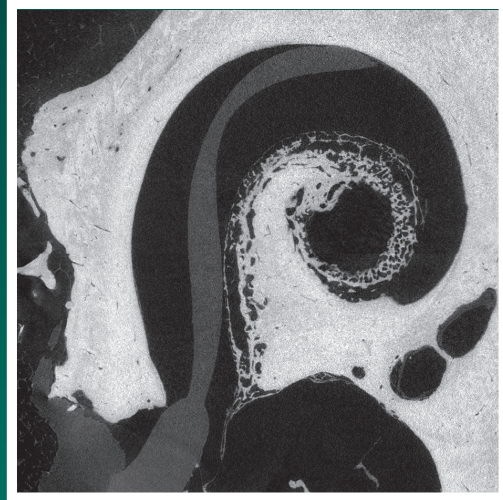
SCANCO MEDICAL

MicroCT Systems and Software

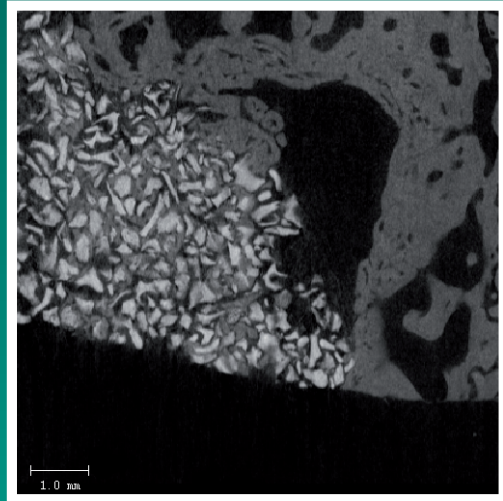




μCT 50, Murine cortical bone, 1 μm



μCT 50, Temporal bone, 6 μm



μCT 100, Grafted bone, 7 μm
© Prof. Pripatnanont, DDS, Prince of Songkhla University

SCANCO - Scanners, Software and Contract Research Services:

SCANCO Medical has been a pioneer in the field of high-resolution computed tomography for more than two decades. We offer a wide range of microCT systems to study the internal structure of practically any material.

These systems are supplied with high-end computing equipment and sophisticated analysis and visualization software to provide the most comprehensive and industry-leading imaging solutions.

SCANCO Medical also offers scanning, analysis and consulting services for a wide range of applications.

Applications:

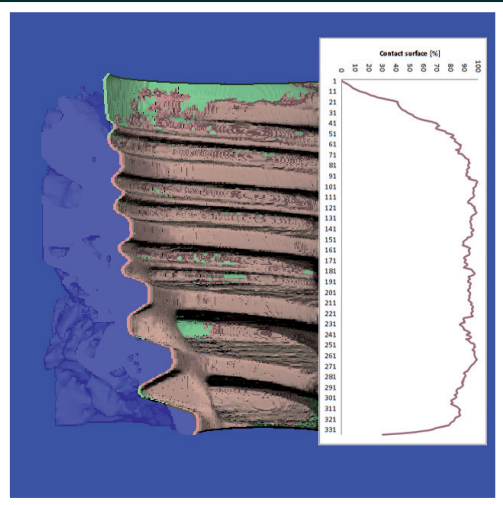
- Bone quality measurements
- Dental research
- Vascular research
- Scaffold characterization
- Implant measurements
- Nondestructive testing
- Imaging of composite materials, polymers and foams
- Imaging of food and biologic material
- Low temperature measurements

High resolution imaging for accurate results:

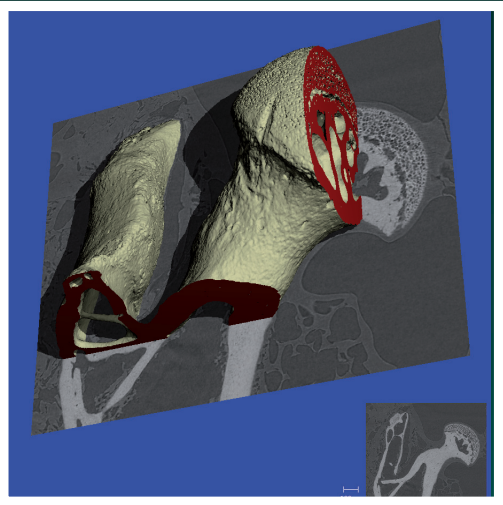
- Non-Destructive analysis
- No sample preparation required
- High throughput (batch mode processing, automatic sample loader)
- Custom system configuration available
- Very fast acquisition, reconstruction and analysis
- Automatic sample changer (μCT 45, 50, 90 and μCT 100/HE)
- Physiological monitoring and gating for vivaCT 40 and 80.

Software

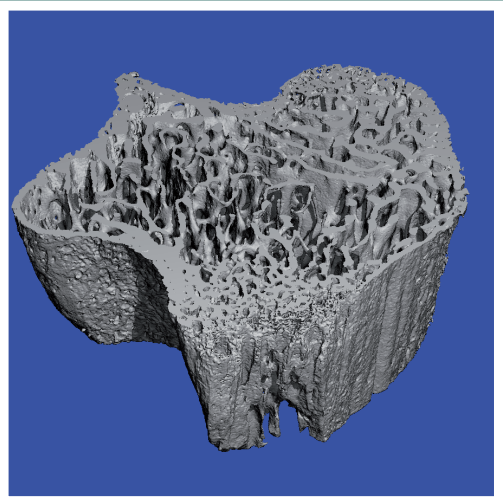
- Bone morphometry and density measurements
- General morphometry: Segmentation, porosity, specific object surface, object volume, local thickness, pore size distribution, pore connectivity
- Contact area calculations
- Visualization in 2D and 3D
- Finite Element Analysis



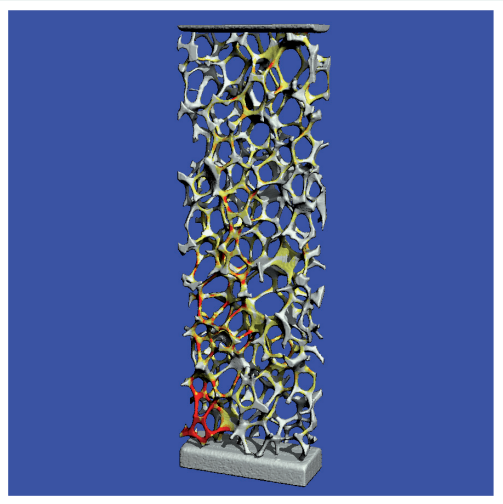
μCT 35, Implant contact surface, 10 μm



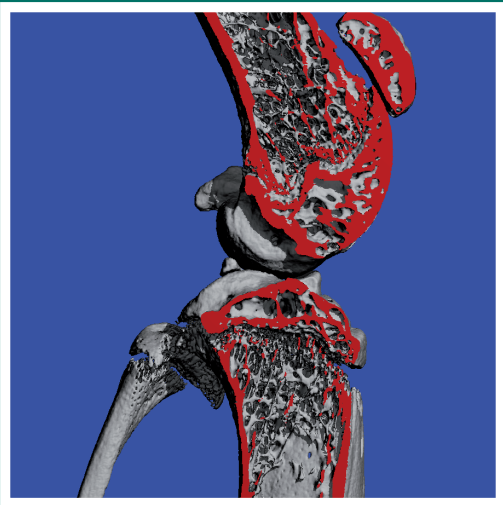
μCT 45, Mouse Femur, 3 μm



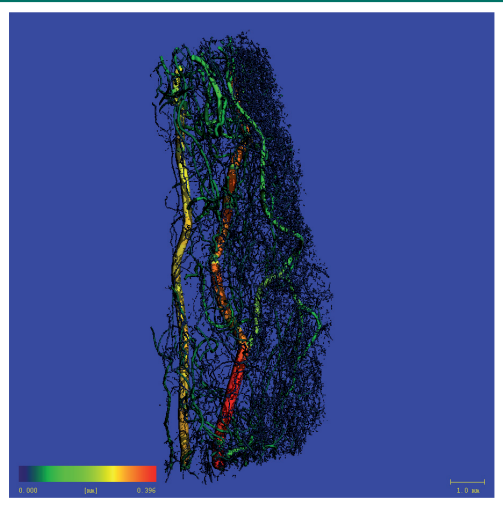
μCT 50, Mouse tibia, 2 μm



μCT 40 Al foam FE Analysis, 6 μm



vivaCT 80, Mouse knee, 5 μm



μCT 100, Skin vascular system, 6 μm